

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

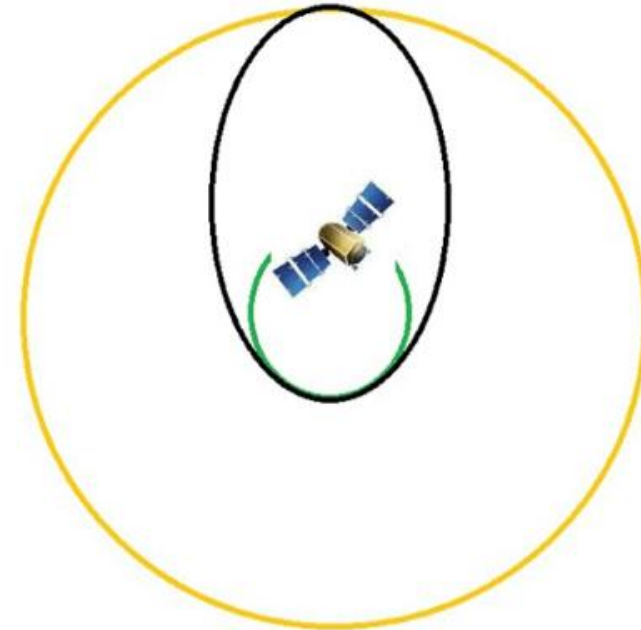
MODULE - 5

INDIAN SPACE RESEARCH ORGANISATION- 3

3. GSLV Mk III



- It is the heaviest launch vehicle made by India. It is also the shortest, so it earned the title 'Fat Boy'.
- GSLV Mk III is designed to carry 4-ton class of satellites into Geosynchronous Transfer Orbit (GTO) and about 8-10 tons to Low Earth Orbit (LEO), which is about twice the capability of GSLV Mk II.



3. GSLV Mk III

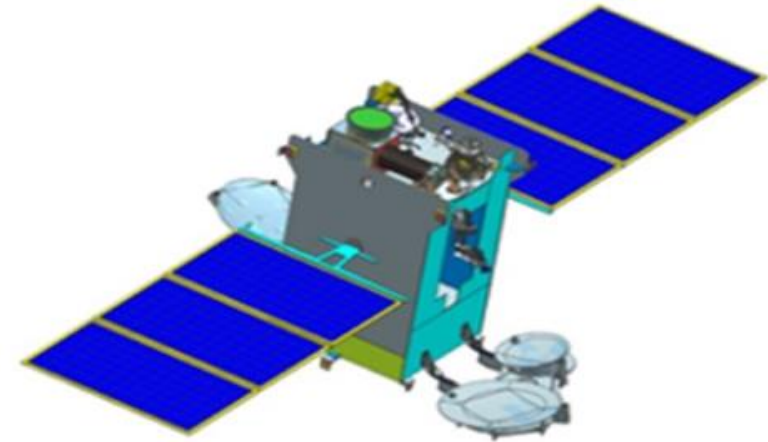
- GSLV Mk III is a three-stage heavy lift launch vehicle developed by ISRO.
- The vehicle has two solid strap-on boosters, a core liquid booster (1st stage) and a cryogenic upper stage powered by Cryogenic Engine (CE)-20 which is
- India's largest cryogenic engine, designed and developed by the Liquid Propulsion Systems Centre.
- The core stage contains two Vikas liquid engines.
- The two strap-on boosters of GSLV Mk III are located on either side of its core liquid booster, they are designated as 'S200', each carries 205 tons of solid propellant and their ignition results in vehicle lift-off.
- In June 2018, the Union Cabinet approved Rs. 4,338 crores to build 10 GSLV Mk-III rockets over a five-year period.



SOME IMPORTANT MISSIONS OF GSLV MK III

1. GSAT-29

- It is a communication satellite developed by ISRO. It was launched on 14th November 2018 by GSLV Mk III rocket.
- It aims at providing high-speed bandwidth to rural areas of Jammu and Kashmir and Northeast India.
- GSAT-29 is the heaviest satellite, weighing 3,423 kg launched by an Indian launch vehicle.
- It also carries a high-resolution camera for imaging. It was placed in geosynchronous transfer orbit (GTO). It will join the INSAT fleet in geostationary orbit.



SOME IMPORTANT MISSIONS OF GSLV MK III

2. CHANDRAYAAN 2

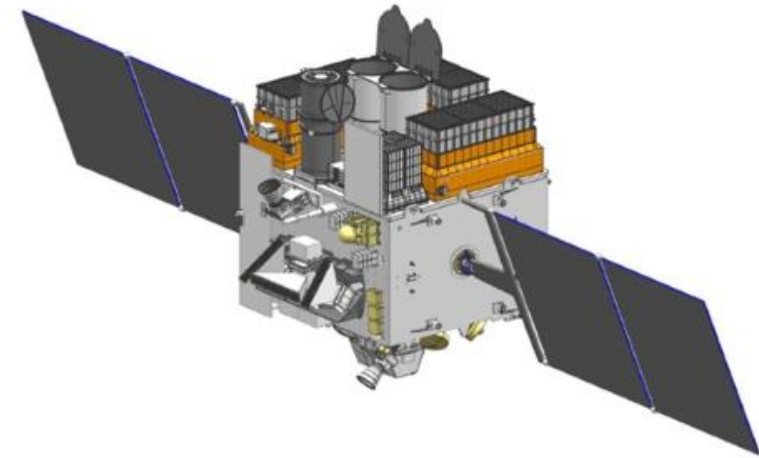
- GSLV Mk III successfully launched Chandrayaan-2 spacecraft on July 22, 2019 into its planned orbit.
- The launch took place from the Satish Dhawan Space Centre SHAR, Sriharikota.
- Chandrayaan-2 mission is a highly complex mission, which represents a significant technological leap compared to the previous missions of ISRO. It comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon.
- The Orbiter was placed in a 100 km lunar polar orbit. Vikram Lander separated from the Orbiter and started descending towards the lunar surface. At an altitude of 2.1 km from the lunar surface, communication from lander to the ground stations was lost.



IMPORTANT ISRO MISSIONS

1. ASTROSAT

- It is India's 1st multi-wave length space observatory.
- Astrosat can make simultaneous observations of an astronomical body in optical, ultra-violet and x-ray radiations.
- It was launched into a 650 km Low Earth Orbit in September 2015, using a PSLV C30 (PSLV XL) rocket.



1. ASTROSAT

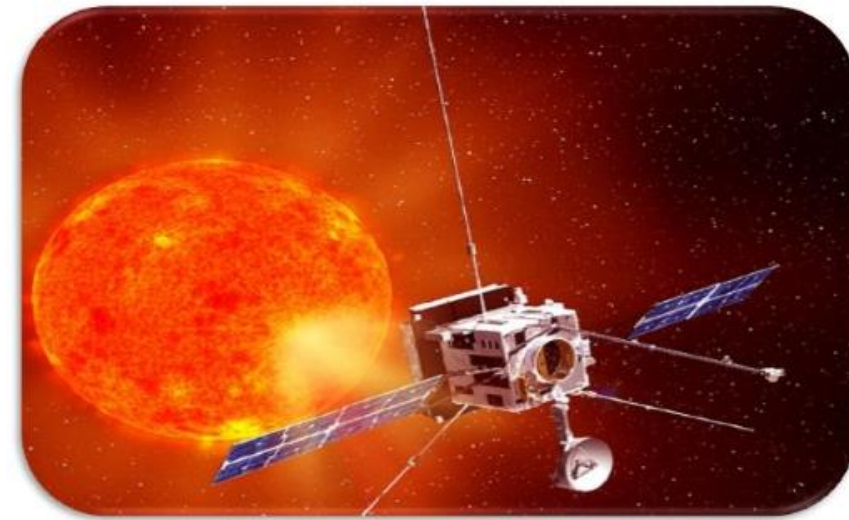
- The objectives of Astrosat are
 - ✓ To understand binary star systems.
 - ✓ To study star-birth regions and nebulas.
 - ✓ To detect high energy x-ray sources.
 - ✓ To perform a survey of the universe in the Ultra Violet region.



- *Recently Astrosat had captured a rare celestial phenomenon known as 'Vampire Star' phenomenon in which a smaller star sucks mass and energy out of a bigger star eventually causing the death of the larger one.*
- *Astrosat has observed more than 800 celestial bodies till date. Data from Astrosat has resulted in more than 100 publications in top scientific journals.*

2. ADITYA L1 MISSION

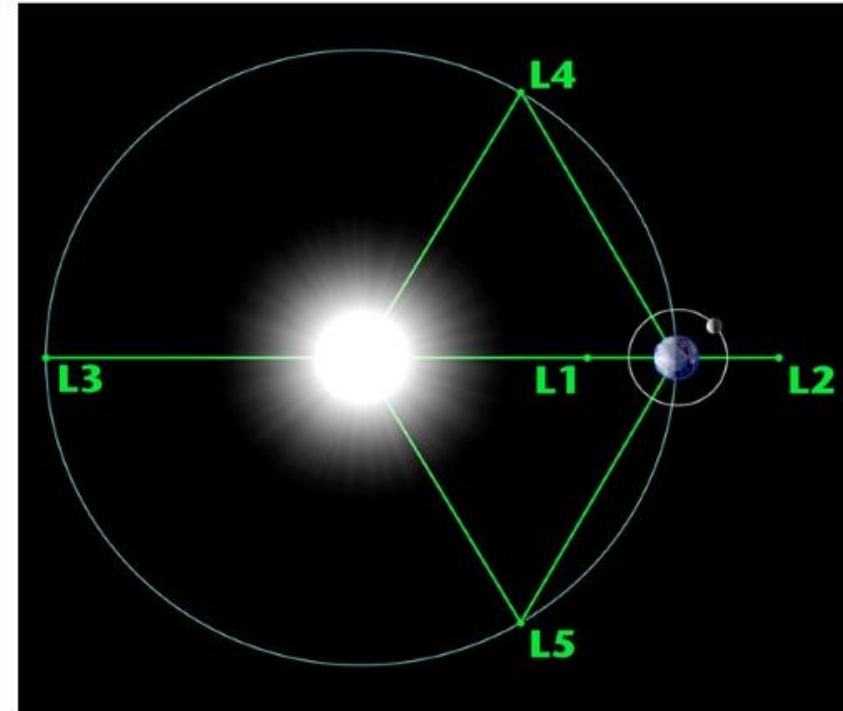
- It is India's 1st mission to study the Sun.
- The original mission was Aditya 1, which was to place a satellite with one instrument (Coronagraph) in 800km Low Earth Orbit.
- But a satellite placed at the Lagrangian point 1 (L1) of the Sun-Earth system has a major advantage of being under less stress from the gravitational pull of the Sun.



IMPORTANT ISRO MISSIONS

LAGRANGIAN POINTS

- These are locations in the space where the gravitational pull of two large bodies roughly balances each other.
- In the Sun-Earth system there are 5 Lagrangian points –
- So the objective of Aditya L1 Mission is to place a satellite at Lagrangian point 1 (L1) which is about 1.5 million kms from Earth. A satellite placed at this point is under less gravitational stress, so it can travel in the orbit at a lesser speed. So, L1 is a good position to monitor the sun.
- The Aditya L1 is expected to be launched in January 2022, using a PSLV XL rocket.



IMPORTANT ISRO MISSIONS

SOLAR CORONA

- Corona is the outer layers of Sun. The Sun's corona extends millions of kilometers into space and is most easily seen during a total solar eclipse.
- Corona has a temperature much higher (10000000 D Celsius) than the temperature of the surface of Sun (5700 D Celsius) known as photosphere. How corona gets heated to such high temperatures is still an unanswered question which Astrosat seeks to answer.



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